

Max Weiner

10/8/2026

ENGR 095

Project Reflection

In the final section of the module, we integrated everything we learned about sensors, circuits, and microcontrollers to design a prototype solution to a real world problem. We began by reviewing the design brief for the project. We analyzed the requirements and constraints to determine what we would make for the project. My group decided on an automatic roof that can extend and retract based on environmental conditions. We chose this project because we felt it was a logical way to include all of the sensors and components required of the prototype, and would be feasible to construct with the materials in the keysight lab that we had access to. After deciding on our concept, we assigned roles and set about developing an initial prototype. I was mostly responsible for the electronics and programming portion of the project. I first tested 3 different types of motors using the Arduino and code developed by Google Gemini to discover what would work best for our project. I settled on a stepper motor, because it would be easy to mount, easily controllable, and provided enough torque. I then set about using what we learned in the last module to wire the sensors, LEDs, and switches into the Arduino with a breadboard. I gave Google Gemini the pin layout of my wiring, and explained what I needed the code to do. With a few iterations, the code worked flawlessly. I made sure to upload each iteration into my github repository, and I helped my other two group members integrate the sensors and motor into the physical roof they designed.

The second portion of the final project was a presentation about our design. I worked with my group and drew on what I learned about how to avoid Death by Powerpoint to create

this presentation. I made sure to keep the amount of text on slides to a minimum, and had engaging graphics. My group worked together to make each slide based on the presentation rubric, and agreed on who would present each slide. While we were practicing presenting, we realized that it was more engaging if we treated the portion of the presentation about our design process as a conversation between the three of us about how we each contributed, so we ended up giving the final presentation in that format.

Throughout the final project, I feel like I was able to put together everything we learned in class. I was able to read the design brief and know what we needed to do to respond to it, I understood the technical aspects of the project, I was able to create a good presentation, and document all of my results. I also think that I learned a lot about how to work in a group. There was sometimes friction about whose idea to go with, or which direction to take, but we were able to resolve any disputes by trusting each other and hearing each other out. There were some times where I was frustrated because we weren't doing things my way, but I trusted the process and the way we did things turned out to be better. I also appreciated how each member of my group was able to contribute their strengths to the project. My other two group members are very mechanically minded, while I really enjoy wiring and creating circuits. I think it led to us having a very successful project.